

Veer Narmad South Gujarat University, Surat



Computer Science and Information Technology Faculty

M.Sc. (Computer Application)(Honours)

902 - Robotic Process Automation (RPA)

UNIT-3

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Syllabus of UNIT 3

Unit 3. Workflow and Selectors

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- 3.2 Variables : DataTypes and Usage, Managing Arguments
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Q.1: Explain Types of Workflows in UiPath

Answer :

In UiPath, there are three main types of workflows: Sequences, Flowcharts, and State Machines. Each serves a different purpose and is useful in various scenarios.

1. Sequences

- Definition: A sequence is a linear flow of activities. It executes each activity one after the other in a specific order.
- Use Cases: Best for simple processes where steps follow a straight line, like data entry or file manipulation.
- Characteristics:
 - Easy to understand and manage.
 - Good for short and straightforward tasks.

2. Flowcharts

- Definition: A flowchart allows for more complex decision-making. It uses a graphical representation where activities can branch out based on conditions.
- Use Cases: Suitable for processes with multiple paths, such as decision-making scenarios or workflows that require looping.
- Characteristics:
 - Provides a visual way to see different paths of execution.
 - Can incorporate activities like decision points (If conditions) and loops (For Each).

3. State Machines

- Definition: A state machine workflow is a more advanced structure that allows the automation to change states based on events.
- Use Cases: Ideal for scenarios where the workflow might need to respond to different events, like monitoring an application status.
- Characteristics:
 - Each state represents a specific stage in the process.
 - Transitions between states are based on triggers, making it flexible for complex applications.

Q.2: Explain Different Variables, Data Types Usage, and How to Manage Arguments

Answer:

Variables

- Definition: Variables are used to store data temporarily during workflow execution.
- Types:
 - String: Used for text (e.g., "Hello, World").
 - Integer: For whole numbers (e.g., 5, 100).
 - Boolean: For true/false values (e.g., true, false).
 - Array: A collection of values (e.g., a list of numbers).

Data Types Usage

- Choosing Data Types: Use appropriate data types based on what you need to store. For example:
 - Use String for names or messages.
 - Use Integer for counts or IDs.
- Example: If you're counting how many times a task is performed, use an Integer variable.

Managing Arguments

- Definition: Arguments are similar to variables but are used to pass data between workflows.
- Types:
 - In Argument: Passes data into a workflow.
 - Out Argument: Sends data back to the calling workflow.
 - In/Out Argument: Can pass data both ways.
- Managing Arguments: When creating arguments, define the direction (In, Out, In/Out) and the type (e.g., String, Integer) for clarity and proper functionality.

Q.3: Explain Data Scraping Giving Appropriate Example

Answer:

Data Scraping

- Definition: Data scraping is the process of extracting structured data from websites or applications.

- Use Case: For example, if you want to gather product prices from an e-commerce site.

Example of Data Scraping

1. Identify Data: Suppose you want to scrape product names and prices from a webpage.

2. Use Data Scraping Wizard: In UiPath, you would use the Data Scraping wizard to select the elements you want to extract.

3. Define Output: The extracted data can be saved in a DataTable or Excel file for further analysis.

Q.4: Explain Recording and Types of Recording

Answer:

Recording

- Definition: Recording in UiPath allows you to capture user actions on the screen and convert them into automated activities.

Types of Recording

1. Basic Recording:

- Captures simple actions like clicks and typing.
- Ideal for straightforward tasks without complex interactions.

2. Desktop Recording:

- Captures detailed actions on a desktop application.
- Suitable for automating more complicated workflows.

3. Web Recording:

- Focused on web applications, capturing actions like navigating and clicking links.
- Designed for web-specific interactions.

4. Citrix Recording:

- Used for automating applications running on a virtual desktop.
- Useful for capturing interactions in remote environments.

Q.5: Describe Recording with Web

Answer:

Web Recording

- Definition: A type of recording specifically designed to automate web applications.

How It Works

1. Start Recording: You initiate web recording mode in UiPath.
2. Capture Actions: As you interact with the web page (clicking buttons, entering data), UiPath captures these actions.
3. Generate Workflow: The recorded actions are converted into a workflow that can be modified and executed later.
4. Benefits: Makes it easy to automate repetitive tasks on web interfaces without manually coding each action.

Q.6: Discuss Manual Recording

Answer:

Manual Recording

- Definition: In manual recording, users specify each action they want to automate instead of having the system capture them automatically.

How It Works

1. User Guidance: Users click on the "Manual Recording" option and then select the actions they wish to include.
2. Adding Activities: Users can manually add activities for actions not easily captured automatically.
3. Use Case: Ideal for workflows with specific conditions or where automation requires customization beyond standard recordings.

Q.7: Explain Concepts of Selectors and Selectors with Wildcards

Answer:

Selectors

- Definition: Selectors are strings that define the user interface elements that an automation will interact with.
- Purpose: They tell UiPath exactly which element on the screen to click, type into, or extract information from.

Selectors with Wildcards

- Wildcards: Special characters that can replace one or more characters in a string.
- Types:
 - Asterisk (*): Represents zero or more characters (e.g., "btn*" matches "btnSubmit").
 - Question Mark (?): Represents a single character (e.g., "btn?" matches "btn1" and "btnA").
- Use Case: If the name of a button changes dynamically, using wildcards in the selector helps ensure that your automation can still find it.

Q.8: Explain Full versus Partial Selectors

Answer:

Full Selectors

- Definition: A full selector describes the entire path to the UI element, including its hierarchy.
- Use Case: Provides precise targeting but can be fragile if any UI structure changes.

Partial Selectors

- Definition: A partial selector includes only the essential attributes needed to identify the element.
- Use Case: More robust against UI changes and can still identify elements as long as some key attributes remain the same.

Q.9: What is UiPath Explorer?

Answer :

UiPath Explorer

- Definition: UiPath Explorer is a tool used to inspect and interact with UI elements on the screen.

- Purpose: Helps users analyze selectors and understand the properties of different UI elements.

Features

- Element Inspection: Users can hover over elements to see their properties and selectors.

- Selector Validation: It can test and validate if a selector accurately identifies the intended UI element.

- Use in Development: A valuable resource for developers to refine selectors for their workflows.

UNIT – 3

MCQ

1. What is the primary function of UiPath Studio?

- A) Manage automation robots
- B) Design and develop automation workflows
- C) Monitor workflow execution
- D) Generate reports
- Correct Answer: B

2. What does UiPath Orchestrator primarily manage?

- A) Data extraction
- B) Robot deployment and scheduling
- C) Workflow design
- D) User interface
- Correct Answer: B

3. Which of the following is a component of UiPath Studio?

- A) Activities Panel
- B) Robot Manager
- C) Control Center
- D) Dashboard
- Correct Answer: A

4. Which version of .NET is required to run UiPath Studio?

- A) .NET Framework 4.5
- B) .NET Framework 4.6.1
- C) .NET Framework 3.5
- D) .NET Core
- Correct Answer: B

5. What is the first step in installing UiPath Studio?

- A) Configure Orchestrator
- B) Download the installer from the UiPath website
- C) Set up a robot
- D) Create a new project
- Correct Answer: B

6. Which panel in UiPath Studio allows users to search for specific activities?

- A) Properties Panel
- B) Output Panel
- C) Activities Panel
- D) Design Panel
- Correct Answer: C

7. What is the purpose of the Ribbon in UiPath Studio?

- A) Displays log messages
- B) Organizes tools and commands
- C) Manages robot schedules
- D) Provides a coding environment
- Correct Answer: B

8. What type of recording captures user interactions with web applications?

- A) Basic Recording
- B) Desktop Recording
- C) Web Recording
- D) Citrix Recording
- Correct Answer: C

9. After recording a workflow, what can you do to refine it?

- A) Publish it immediately
- B) Edit activities in the Design Panel
- C) Delete all activities
- D) Set it to run unattended
- Correct Answer: B

10. Which activity is commonly used to input data into a text field?

- A) Click
- B) Type Into
- C) Assign
- D) Log Message
- Correct Answer: B

11. What is the purpose of variables in UiPath?

- A) Store user interface elements
- B) Hold data values during automation
- C) Control workflow execution
- D) Generate reports
- Correct Answer: B

12. What does the Try-Catch activity do in UiPath?

- A) Terminates the workflow
- B) Attempts to execute a block of code and handles exceptions
- C) Logs all activities
- D) Runs the workflow in debug mode
- Correct Answer: B

13. What panel provides real-time logs of the workflow execution?

- A) Properties Panel
- B) Output Panel
- C) Activities Panel

- D) Control Panel
- Correct Answer: B

14. Which tab on the Ribbon would you use to publish your workflow?

- A) Home
- B) Design
- C) Debug
- D) Publish
- Correct Answer: D

15. What can you find in the Debug tab of the Ribbon?

- A) Workflow settings
- B) Execution control options
- C) Activity properties
- D) Activity library
- Correct Answer: B

16. What is the primary use of the Universal Search Box in UiPath Studio?

- A) Search for external libraries
- B) Find specific activities or components
- C) Run the workflow
- D) Access project settings
- Correct Answer: B

17. Can the Universal Search Box help in locating variables?

- A) Yes
- B) No
- C) Only if they are global
- D) Only if they are in the Library Panel
- Correct Answer: A

18. Which panel is used to organize and access reusable components?

- A) Activities Panel
- B) Library Panel
- C) Properties Panel
- D) Output Panel
- Correct Answer: B

19. What does the Project Panel display?

- A) All activities available in UiPath
- B) The hierarchy of files and resources in the current project
- C) Workflow execution logs
- D) Active robots
- Correct Answer: B

20. What information is primarily shown in the Properties Panel?

- A) Execution logs
- B) Settings and configurations for the selected activity
- C) Available activities
- D) Workflow hierarchy
- Correct Answer: B

21. What does the Outline Panel help you manage?

- A) Activity settings
- B) Workflow structure and navigation
- C) Robot schedules
- D) Installed libraries
- Correct Answer: B

22. What does the Control Panel allow you to do during debugging?

- A) Set breakpoints
- B) Control the execution of the workflow
- C) Search for activities
- D) View execution logs
- Correct Answer: B

23. Which activity would you use to scrape data from a website?

- A) Data Scraping
- B) Read Range
- C) Click
- D) Type Into
- Correct Answer: A

24. What type of robot executes tasks without human intervention?

- A) Attended Robot
- B) Unattended Robot
- C) Assistant Robot
- D) Hybrid Robot
- Correct Answer: B

25. Which panel would you primarily use to create complex workflows?

- A) Activities Panel
- B) Design Panel
- C) Properties Panel
- D) Output Panel
- Correct Answer: B

26. What is the function of the Write Line activity?

- A) To log information to the Output Panel
- B) To write data to an external file

- C) To display a message box
- D) To input data into a form
- Correct Answer: A

27. Which of the following is a key benefit of using UiPath Orchestrator?

- A) Designing workflows
- B) Monitoring robot performance and scheduling tasks
- C) Recording user interactions
- D) Accessing the Activities Panel
- Correct Answer: B

28. What does a flowchart in UiPath represent?

- A) A linear sequence of activities
- B) Conditional branching and loops
- C) Only data manipulations
- D) A list of activities
- Correct Answer: B

29. Which activity is used to click a UI element?

- A) Type Into
- B) Click
- C) Select Item
- D) Get Text
- Correct Answer: B

30. What is the purpose of the Selector in UiPath activities?

- A) Define the workflow structure
- B) Identify UI elements for automation
- C) Store variable values
- D) Manage project settings
- Correct Answer: B